

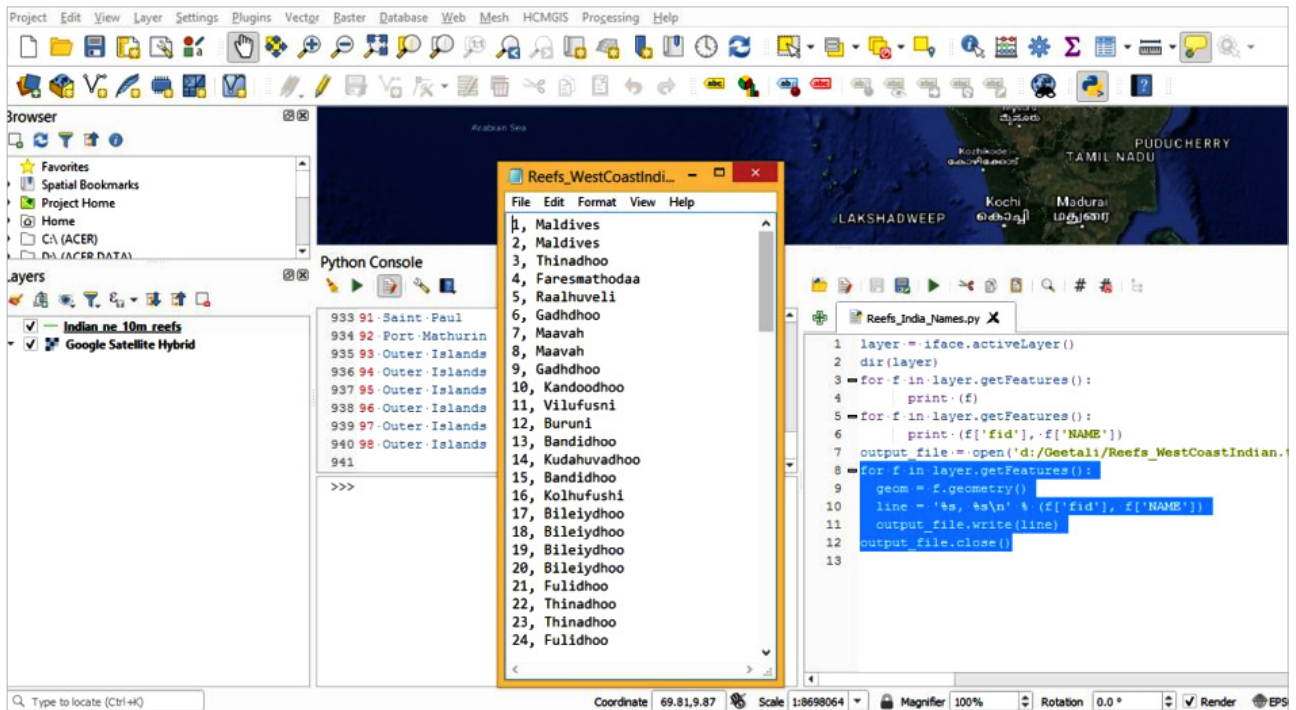


Figure 9: Names of the 98 reefs on the western coast of India

A wonderful resource for learning common task execution is to download the existing plugins from the plugin repository. QGIS provides an integrated Python console for scripting. It can be accessed from the *Plugins ► Python Console* menu.

QGIS essentially works as a layer based approach, and hence it becomes vital to obtain the currently selected layer in the layer list on the top. The ID visibility is optional. If the current layer happens to be a vector layer, the features are the most vital components and so is the feature count. In the QGIS environment, the *iface* variable, an instance of *QgisInterface*, allows the user to access the map canvas, menus, toolbars and other parts of the QGIS application.

The *iface.activeLayer()* method gives us the currently selected layer.

```

>>>layer = iface.activeLayer()
>>>dir(layer)

for f in layer.getFeatures():
    print (f)

```

Executing the code in the console will provide the output, as shown in Figure 8.

```

for f in layer.getFeatures():
    print (f['fid'], f['NAME'])
output_file = open('d:/Geetali/Reefs_WestCoastIndian.txt', 'w')

```

This will generate a *.txt* document at the desired location with the desired name.

Finally we will execute the code

below to display the names of the reefs

```

for f in layer.getFeatures():
    geom = f.geometry()
    line = '%s, %s\n' % (f['fid'], f['NAME'])
    output_file.write(line)
output_file.close()

```

Figure 9 shows that after complete execution of the Python code, we have successfully loaded the names of the 98 reefs on the western coast of India. 

## References

- [1] [https://docs.qgis.org/3.22/en/docs/pyqgis\\_developer\\_cookbook/intro.html#scripting-in-the-python-console](https://docs.qgis.org/3.22/en/docs/pyqgis_developer_cookbook/intro.html#scripting-in-the-python-console)
- [2] Made with Natural Earth. Free vector and raster map data @ [naturalearthdata.com](http://naturalearthdata.com).
- [3] <https://products.aspose.app/gis/en/conversion/shapefile-to-kml>

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